When Does Write-Time Augmentation Help Retrieval? A Divergence Diagnostic for RAG Pipelines

Jeffrey Chang

April 2026

Abstract

Write-time document augmentation — adding LLM-generated questions (docT5query), structured metadata, or multiple chunk representations — is a popular retrieval-pipeline lever, but published gains (+3–10 pp on single corpora) replicate inconsistently. We ask *when* augmentation helps and propose a free, pre-deployment diagnostic: estimate the mean Jaccard divergence between ~ 50 held-out (query, target) pairs and act conditionally on the result. We test the diagnostic and its conditional recommendations on five corpora: a stratified enterprise benchmark of 160 queries (with an 80-query held-out split), three BEIR test sets (SciFact, NFCorpus, FiQA), LoTTE-writing-forum, and HotpotQA-500 multi-hop. Three findings support the diagnostic. (i) **Per-query rank improvement is a detectable monotone function of query-target Jaccard divergence within a corpus**: regression slopes are positive and significant on the corpus where augmentation works (enterprise: β=+1.41, p=0.008, R^2=0.05), and indistinguishable from zero on corpora where it does not (SciFact: β=+0.24, p=0.62, R^2<0.01). The same regression separates the "augmentation-friendly" from "augmentation-neutral" regime that prior work has reported as cross-corpus heterogeneity. (ii) **RODS, a six-field schema we introduce, and docT5query are statistically indistinguishable on R@5 in-distribution** (p=0.52), and **both R@5 lifts collapse on a held-out split** (p=0.73). RODS-M4 retains a held-out MRR@10 lift (+5.0 pp, raw p=0.011), suggestive but borderline under multiple-comparisons correction. (iii) **Cross-encoder reranking absorbs the single-chunk schema gain** (p=0.67) **but not the multi-vector advantage** (+6.5 pp R@5 post-rerank, p<0.001; +8.1 pp held-out MRR@10, p=0.001); a 4× stronger embedder (`all-mpnet-base-v2`) flips the comparison, widening the single-chunk schema gap and erasing the quad-vector advantage. **Under Holm–Bonferroni correction, only the multi-vector MRR@10 lift survives at family-wise** α=0.05; we therefore present multi-vector as the robust positive result and the single-chunk schema gains as suggestive. The diagnostic resolves much of the cross-corpus heterogeneity reported in prior work: format design, multi-vector indexing, reranking, and embedder scale are complementary levers whose relative payoff is set by query–document divergence and embedder budget. We release the diagnostic harness, the held-out benchmark, and per-query rank traces.

1 Introduction

Consider an internal wiki where the section headed *"Decision: ADR-005"* contains the line *"We standardize on Python 3.12 for the backend"* and a metadata block reading `Status: accepted`. A user types *"what is our standardized backend Python version?"*. None of the query's content tokens appear in the heading, and only one appears in the prose; under a default heading-aware RAG pipeline this section ranks fifth, behind sections that share token *Python* but discuss something else. Write-time augmentation — adding an explicit `Summary`, `Type`, or `Entities` field above the prose — closes this gap. Whether it helps is the question this paper studies.

More generally: a RAG pipeline is a sequence of fixed components — a chunker, an embedder, a retriever, optionally a reranker, and a language model — each with its own literature of optimisations. *Write-time* *document augmentation* (modifying what is written into the index, rather than how it is retrieved) has produced single-corpus gains of $+3$–10 pp R@5 across this literature (docT5query, HyDE, ad-hoc per-section schemas, multi-vector decompositions, Contextual Retrieval), but the gains replicate inconsistently across corpora and rarely survive a held-out split or a cross-encoder reranker. The field has no predictive theory of *when* write-time augmentation is worth paying for.

Our proposal: a pre-deployment diagnostic. On ~ 50 held-out (query, target) pairs, regress per-query rank improvement (relative to a heading-aware baseline) on Jaccard divergence between query and target tokens, and read off slope β and significance. A positive, significant β predicts that write-time augmentation will lift retrieval on this corpus; β indistinguishable from zero predicts that it will not. The diagnostic is parameter-free, uses no LLM and no embedder, and runs in seconds.

RODS as a worked instance. To give the diagnostic a concrete object we release *RODS* (Retrieval-Oriented Document Structuring), a per-section schema of up to six typed fields with an incremental ablation chain $M_0 \to M_7$; the held-out check (§6.2) selects the deployable cut.

Branched recommendation. Beyond the go/no-go test, our cross-encoder rerank and embedder-swap experiments yield a recommendation that branches on embedder budget rather than picking a single "best format" (§9).

Contributions. We make three contributions.

1. **A pre-deployment divergence diagnostic for write-time augmentation, validated on five corpora.** We regress per-query rank improvement on Jaccard divergence $\mathrm{div}(q, d)=1-|T(q) \cap T(d)|/|T(q)|$ and report slope β, intercept α, and R^2. The diagnostic is positive on corpora where augmentation works ($\beta_{\mathrm{RODS-M6}}=+1.41$, $p=0.008$ on enterprise) and indistinguishable from zero where it does not (SciFact: $\beta=+0.24$, $p=0.62$); R^2 is small everywhere, but the sign and significance separate the regimes cleanly. Across all 5 corpora $\times 4$ formats, the diagnostic's verdict agrees with the observed augmentation outcome on $16/20$ cells with no false positives (Table 11). Numbers are reported with 95% bootstrap CIs, paired-bootstrap p-values, an 80-query held-out split, and a cross-encoder rerank / large-embedder ablation matrix.
2. **An open held-out enterprise benchmark.** 160 stratified queries across 8 categories, 12 documents, 72 sections; the 80 held-out queries were authored without reference to the schema generator. The benchmark is small enough that Recall@5 partially saturates, and we report per-category breakdowns to make this explicit.
3. **RODS, an open six-field schema with ablation chain.** Single-chunk, ~ 70 tokens per section. We release the schema, the eight-step ablation, the held-out queries, and the diagnostic harness. The deployable cut emerges from the held-out test in §6.2.

2 Related Work

Write-time document augmentation. Several methods modify the indexed text rather than the retriever: *docTTTTTquery* [Nogueira and Lin, 2019] appends K predicted queries to each document and remains the strongest reproducible single-chunk baseline; *Contextual Retrieval* [Anthropic, 2024] prepends chunk-specific LLM-generated context (50–100 tokens) before embedding and BM25 indexing, reporting a 35% reduction in failed retrievals on a multi-domain set (49% when paired with contextual BM25; 67% adding a Cohere reranker). RODS sits in the same family — write-time text added before the body to give the embedder more surface to match on — but with a typed, ablated schema rather than free-form generated context, which lets us attribute gains to specific fields. *HyDE* [Gao et al., 2023a] is the query-side counterpart: it expands the query at retrieval time with a hypothetical answer generated by an LLM. Our HyDE implementation uses a template-based cache (we lacked query-time LLM access) and is reported as *exploratory*. Various wikis and RAG toolkits expose ad-hoc schemas (heading metadata, summary fields, keyword tags) without controlled comparison; our contribution is the diagnostic that predicts *when* any of these methods will help, not a new schema in isolation.

Multi-vector retrieval. ColBERT [Khattab and Zaharia, 2020] indexes one vector per token and retrieves via late interaction; follow-ups such as ColBERTv2 [Santhanam et al., 2022] compress the index. Our "three-vector" / "quad-vector" formats are sub-section decompositions evaluated by a single bi-encoder rather than a token-level model, but share the underlying intuition that *candidate-pool diversity* is a different lever from *per-chunk content quality*. Our reranker-absorption result (§6.6) gives this distinction empirical content: a cross-encoder cannot recover at query time what a richer candidate pool exposes.

Hybrid retrieval and rank fusion. Reciprocal Rank Fusion [Cormack et al., 2009] is a parameter-free combiner of ranked lists that we use throughout. BM25+dense hybrids are now standard in IR practice; we report all numbers under hybrid retrieval to match production deployments.

Cross-encoder reranking. The ms-marco-MiniLM cross-encoder family [Nogueira and Cho, 2019, Reimers and Gurevych, 2019] is the standard end-of-pipeline reranker. Prior work documents reranker gains on top of bi-encoders [Nogueira and Cho, 2019] but rarely tests whether write-time augmentation contributes *after* a reranker. We add this test (§6.6) and find that single-chunk schema gains are absorbed but multi-vector candidate-pool gains are not.

RAG diagnostics and retrieval quality. Thakur et al. [2021] document large cross-corpus heterogeneity in retrieval gains; Gao et al. [2023b] survey RAG components but do not predict which interventions are worth paying for on a given workload. We propose a corpus-level pre-deployment diagnostic (Jaccard divergence of held-out pairs) that complements these surveys with an actionable, reproducible test.

Retrieval evaluation rigour. BEIR [Thakur et al., 2021] and BEIR-style splits are the standard external-validity check; LoTTE [Santhanam et al., 2022] contributes long-tail forum and STEM question-answering data; HotpotQA [Yang et al., 2018] contributes multi-hop. We run all three families plus a synthetic enterprise benchmark with held-out authoring, paired-bootstrap significance, and bootstrap CIs. Held-out splits authored separately from the schema generator are uncommon in write-time-augmentation papers; we treat the held-out comparison (§6.2) as the central credibility check.

3 Background and Setup

Retrieval pipeline. A document corpus $\mathcal{D} = \{d_i\}$ is partitioned into sections $\{s_{i,j}\}$ at heading boundaries. Each section is mapped by a format F to one or more chunks $\{c_{i,j,k}\} = F(s_{i,j})$; in this work we focus on single-chunk formats ($k = 1$) except where noted. A retriever R embeds the chunks and returns, for a query q, the top-K chunks ranked by similarity. A section is *retrieved at rank r* if rank r is the smallest position at which any chunk derived from that section appears.

Retrievers. We use three retrievers throughout. **TF-IDF** is a sparse baseline with unigram+bigram features and sublinear term frequency. **Dense** embeds chunks via the `all-MiniLM-L6-v2` sentence transformer (384-dim) under cosine similarity; §6.7 swaps in `all-mpnet-base-v2` (768-dim) as a robustness check. **Hybrid** runs both and combines per-chunk ranks via Reciprocal Rank Fusion (RRF):

$$\mathrm{RRF}(c) \;=\; \frac{1}{k + r_{\mathrm{tfidf}}(c)} + \frac{1}{k + r_{\mathrm{dense}}(c)}, \qquad k = 60.$$

RRF is parameter-free, robust to score-distribution differences between retrievers, and adds at most one TF-IDF cosine and a sort to query latency. Throughout the paper, the headline numbers are *hybrid* unless we explicitly compare retrievers.

Metrics. For single-hop retrieval we report Recall@K for $K \in \{3, 5, 10, 20\}$, MRR@10, and nDCG@10. For multi-hop retrieval we report `both@K`, the fraction of queries where both ground-truth sections appear in the top-K retrieved set. We also report cost per format: chunks per section (memory proxy), tokens per section (index size proxy), and wall-clock query latency in milliseconds.

Significance testing and confidence intervals. Throughout the paper, 95% confidence intervals are non-parametric percentile bootstrap CIs over $1{,}000$ resamples of the per-query metric. For paired comparisons of two formats A and B on the same query set $\{q_i\}_{i=1}^{n}$, we compute the per-query difference $\delta_i = m_B(q_i) - m_A(q_i)$ for metric $m \in \{R@5, MRR@10\}$, and report (i) the observed mean $\bar{\delta} = n^{-1}\sum \delta_i$ and (ii) a two-sided paired-bootstrap p-value $p = 2 \cdot \min\bigl(\Pr_b[\bar{\delta}_b^* \le 0], \Pr_b[\bar{\delta}_b^* \ge 0]\bigr)$ where $\bar{\delta}_b^*$ is the mean of δ in bootstrap resample b, over B=2000 resamples. For per-bin divergence regressions we use a 2000-fold permutation test of the slope. We mark p<0.05 with $**$.

4 RODS: A Lightweight Retrieval-Oriented Schema

RODS adds up to six typed fields above each section's body, in human- readable form: a one-sentence answer-first `Summary`; a `Type` label drawn from twelve categories (*decision, status_update, email, note, recipe, policy, guide, design_pattern, data_structure, system_concept, testing_concept, topic*); a comma-separated `Entities` list of capitalised noun phrases; an `Aliases` list of synonyms; a small `Status / Date / Owner` block; and a `Related` cross-reference field. The schema is retriever-agnostic — field names are passed through unmodified to whatever sparse, dense, or hybrid retriever consumes the chunks. Each section remains a single chunk in the index; §5.2 contrasts this with multi-vector formats that spend 3–8× more memory.

We evaluate an eight-step ablation chain $M_0 \to M_7$ adding fields one at a time (§5.3); the held-out check in §6.2 selects the deployable cut. Schema fields are produced at corpus-build time, either by direct authoring (the synthetic enterprise corpus) or by heuristic extractors plus a single LLM question-tag pass (BEIR / HotpotQA), at ~ 1 LLM call per section amortised across all queries; generation-cost trade-offs are discussed in §8. Figure 1 shows the maximal schema with the recommended M4 cut highlighted.

```
## Decision:  use Postgres for the Atlas event store
Summary:  We adopt Postgres; familiarity beats Cassandra at our scale.
Type:  decision
Entities:  Postgres, Atlas, Cassandra, event store, ACID, replication
--- ablated; not in recommended cut ---
Aliases:  ADR-001 reversal context; database choice; pricing OLTP
Status:  accepted
Date:  2026-03-04
Owner:  Sara Chen, Marcus
We considered Cassandra for write scale but the team's prior
production experience and the existing migration tooling favour
Postgres.  ...
```

Figure 1: A section under RODS. The four fields shown in black — `Summary`, `Type`, `Entities` above the prose, plus the heading — are the deployable cut RODS-M4 that survives held-out testing (§6.2). The greyed-out fields (`Aliases`, `Status/Date/Owner`; M5–M6) are retained in the maximal schema RODS-M6 we ablate against, but they do not add held-out signal in our evaluation. `Related` (M7) regresses and is omitted.

5 Experimental Setup

5.1 Corpora

Enterprise-knowledge benchmark (synthetic). Twelve documents, 72 sections, 212 question-answer pairs across software engineering, recipes, legal/policy, travel, meeting notes, mock email threads, project-status journals, architecture decision records, and personal notebooks. Stress paraphrases for each query (terse, formal, jargon variants, 848 stress queries total). Eighty additional queries are stratified into eight categories targeting distinct retrieval challenges: fact lookup, decision lookup, rationale, multi-hop, temporal-superseded, procedural, ambiguous entity, and open-ended summary.

BEIR subsets. Three small BEIR test sets, accessed via the standard BEIR distribution. **SciFact** (5,183 docs, 300 test queries) verifies scientific claims. **NFCorpus** (3,633 docs, 323 queries) is a medical information-retrieval set with an average of 38 relevant docs per query. **FiQA** (corpus capped to 5,000 docs from the original 57,638 to fit a small index, all qrel-referenced docs preserved; 648 test queries) is a financial QA set.

HotpotQA. A 500-query slice of the HotpotQA `distractor` validation split, with 2,500 unique context documents (always including all qrel-referenced supporting documents). All queries are multi-hop: each requires synthesising information from exactly two distinct supporting documents.

5.2 Baselines

We compare RODS against four single-chunk baselines, organised by strength:

L0 — Fixed-size chunks Sections split by token budget (256 tokens per chunk, 32 overlap), ignoring heading boundaries. The standard naive RAG baseline.

L1 — Heading-aware chunks Sections split at markdown heading boundaries; each section becomes one chunk. This is what most wiki-ingest pipelines do today.

L2 — Semantic (sentence-boundary) chunks Sections split into $\sim$3-sentence groups with the section title prepended. This approximates LLM-based semantic chunking without an LLM call.

L3 — LLM-summary chunks Each section replaced by a one-sentence summary plus two concept tags. Tests whether a tightly summarised chunk can match RODS at retrieval time.

Multi-vector formats (reference, 3–8$\times$ memory). A separate family of write-time augmentations indexes *multiple* chunks per section instead of one. We define and evaluate two:

- **Three-vector**: each section produces three independent chunks: c_{meta} (the schema fields only: `Summary`, `Type`, `Entities`, etc.), c_{anc} (the ancestor-content vector: heading-path ancestors plus the section's first sentence), and c_{inline} (the section's prose, unchanged). Each chunk is indexed independently; at retrieval, ranks come from the same hybrid retriever (TF-IDF + dense, RRF) over the union of *all* chunks. We deduplicate to one row per section post-retrieval by keeping each section's best-ranked chunk.
- **Quad-vector**: three-vector plus a fourth chunk c_{syn} that contains `Aliases` expanded into a flat synonym list (one chunk per section dedicated to query paraphrases). Indexed and deduplicated the same way as three-vector.

A six-chunk sub-section variant (one chunk per Q-tag) is reported in the appendix for completeness. These multi-vector formats are *not* part of RODS' single-chunk hypothesis; we include them to quantify the cost-quality trade-off at 3–4$\times$ the index memory and embedding cost.

Micro-augmentation baselines (BEIR only). On BEIR we additionally compare three minimal single-chunk augmentations to probe whether smaller-than-RODS interventions help: $\mathbf{V_{rep}}$ (section title repeated twice at the top of the chunk), $\mathbf{V_{kw}}$ (a flat inline keyword list prepended), and $\mathbf{V_{qe}}$ (a single claim-style LLM-generated query expansion prepended). These are deliberately weaker than RODS; they exist to bracket the noise floor of single-chunk write-time augmentation on aligned-vocabulary IR.

5.3 Schema ablation

We evaluate eight incremental levels, each adding one field: **M0** plain prose (markdown stripped); **M1** headings only (= L1); **M2** + Summary; **M3** + Type; **M4** + Entities; **M5** + Aliases; **M6** + Status / Date / Owner; **M7** + Related. The marginal contribution of a field is the delta between consecutive levels.

6 Results

Scope. §6.1–6.2 run on a synthetic enterprise corpus of 12 documents and 72 sections that we constructed to stress eight retrieval categories. We treat its in-distribution numbers as a tight upper bound: five of eight categories saturate at R@5=1.000, so any single-chunk lift comes from ≤ 3 flipped queries in non-saturated buckets. The held-out queries (§6.2), BEIR / LoTTE / HotpotQA results (§6.3–§6.4), the cross-encoder rerank test (§6.6), and the embedder swap (§6.7) are the external-validity checks. We mark which findings survive each of those checks at the end of §6.7.

6.1 Enterprise-knowledge benchmark: in-distribution result

Table 1 reports the in-distribution result on the 80-query stratified enterprise benchmark with 95% bootstrap CIs (1,000 resamples).

Table 1: In-distribution enterprise benchmark, 80 queries across 8 categories, hybrid retrieval. Numbers in [] are 95% bootstrap CIs. Single-chunk RODS variants and docT5query are statistically indistinguishable here (p=0.52, Table 3); the held-out check (§6.2) is what selects the deployable cut. Multi-vector formats lead at 3–4× chunks.

Format	R@5 [95% CI]	MRR@10 [95% CI]	Chunks	Notes
Structural baselines (single-chunk)				
L0 fixed-chunks	0.916 [0.846,0.971]	0.865 [0.797,0.924]	1×	weakest
L1 heading-aware	0.916 [0.846,0.971]	0.865 [0.797,0.924]	1×	current standard
L2 semantic	0.929 [0.872,0.978]	0.882 [0.821,0.937]	∼3×	sentence-bnd
L3 LLM-summary	0.868 [0.787,0.937]	0.791 [0.713,0.864]	1×	body removed
Write-time augmentation baselines (single-chunk)				
docT5query	0.963 [0.917,0.996]	0.922 [0.870,0.964]	1×	5 pred. queries
HyDE (retrieval-time)	0.835 [0.756,0.908]	0.756 [0.680,0.833]	1×	query expansion
RODS schema ablation (single-chunk)				
RODS-M1 (headings)	0.935 [0.879,0.978]	0.896 [0.835,0.948]	1×	= L1
RODS-M2 (+Summary)	0.939 [0.891,0.978]	0.914 [0.857,0.961]	1×	
RODS-M4 (+Entities)	0.945 [0.891,0.983]	0.902 [0.845,0.955]	1×	
RODS-M6 (full)	**0.953** [0.905,0.989]	0.897 [0.839,0.948]	1×	
RODS-M7 (+Related)	0.929 [0.873,0.973]	0.898 [0.844,0.946]	1×	R@5 regresses
Multi-chunk reference (3–4× memory cost)				
Three-vector	0.985 [0.964,1.000]	0.915 [0.865,0.959]	3×	
Quad-vector	0.973 [0.941,0.997]	0.916 [0.865,0.960]	4×	

Interpretation. The maximal schema RODS-M6 reaches Recall@5 = 0.953 [0.905, 0.989], +3.7 pp over L1 (p=0.040, Table 3). However adjacent M-level CIs overlap substantially, so within-RODS deltas are not individually significant; we discuss the ablation as a monotone trend rather than per-step significance. **docT5query** — a simpler write-time baseline that appends 5 predicted queries per section — is statistically indistinguishable from RODS-M6 here (p=0.52). We do not claim RODS beats docT5query on this n=80 synthetic in-distribution set; the credibility check is the held-out split below (§6.2). All in-distribution numbers in this table should be read as a tight upper bound: five of eight query categories saturate at R@5=1.000 (Table 4), so leaderboard "wins" here come from at most 2–3 queries flipping in the open-ended-summary and ambiguous-entity buckets.

6.2 Held-out validation and significance testing

Because the 80 in-distribution queries were authored alongside the synthetic corpus, the schema generator and the query author shared ground-truth knowledge. To test whether RODS gains generalise we authored a separate *held-out* set of 80 queries in the same eight categories, written without reference to the schema generator's outputs and using deliberate paraphrase, negation, and off-vocabulary phrasings.

Scope of the held-out check. The held-out test in this subsection applies only to our synthetic enterprise corpus. BEIR (SciFact / NFCorpus / FiQA), LoTTE-writing-forum, and HotpotQA-500 all use their own published query sets, which were authored independently of *any* schema generator and are therefore intrinsically out-of-distribution for our schema — but they are not a held-out check on the same underlying corpus. We treat the held-out enterprise split as the strongest credibility check of the schema's transfer to fresh queries on

Table 2: Held-out enterprise benchmark, 80 fresh queries (same 8 categories; constructed to avoid lexical overlap with corpus vocabulary). Hybrid retrieval. Numbers in [ ] are 95% bootstrap CIs. **The R@5 lift disappears; the MRR@10 lift survives.** †HyDE in-distribution (R@5=0.835, see Table 1) regressed below the L1 baseline because our hypothetical-answer cache is template-generated rather than LLM-generated; rather than report a misleading held-out number for an under-specified implementation we omit the row, and discuss the limitation in §8.

Format	R@5 [95% CI]	MRR@10 [95% CI]	Chunks
L1 heading-aware	0.908 [0.845, 0.959]	0.808 [0.733, 0.876]	1×
L2 semantic	0.895 [0.820, 0.954]	0.779 [0.703, 0.845]	∼3×
L3 LLM-summary	0.759 [0.666, 0.846]	0.648 [0.560, 0.736]	1×
docT5query	0.911 [0.851, 0.965]	0.812 [0.741, 0.875]	1×
HyDE †	–	–	1×
RODS-M2	0.900 [0.833, 0.963]	0.841 [0.767, 0.906]	1×
RODS-M4 (+Entities)	0.914 [0.855, 0.969]	**0.858** [0.794, 0.920]	1×
RODS-M6 (full)	0.904 [0.838, 0.961]	0.839 [0.773, 0.896]	1×
RODS-M7 (+Related)	0.923 [0.867, 0.969]	0.863 [0.803, 0.918]	1×
Three-vector	0.942 [0.894, 0.981]	0.889 [0.829, 0.941]	3×
Quad-vector	**0.935** [0.880, 0.979]	**0.893** [0.835, 0.945]	4×

a known corpus, and the external-corpus evaluations (§6.3–§6.4) as the credibility check on transfer to fresh corpora. Both checks are reported; both are needed.

The pattern shifts substantially relative to in-distribution. The single-chunk R@5 lift collapses for every write-time augmentation: RODS-M6 (0.904), RODS-M4 (0.914), and docT5query (0.911) all sit within noise of L1 (0.908). MRR@10, however, separates the methods. **RODS-M4 is the schema variant whose held-out gain survives**: +5.0 pp MRR@10 over L1 (p=0.011). RODS-M5 and M6 do not add held-out signal beyond M4. docT5query's in-distribution MRR lift evaporates on held-out (+0.003, p=0.88). Multi-vector decomposition is the largest held-out gain we measure: +8.1 pp MRR@10 (p=0.001) at 3× chunk cost.

RODS-M4's MRR@10 gain is the only single-chunk lift that survives this test — plausibly because semantically-grouped typed fields cluster the embedding more coherently than five queries appended at random. **We therefore recommend RODS-M4 as the deployable schema variant**; M5–M7 are ablation context. The full survival picture across all robustness checks is in Table 10 (§6.8).

Per-category breakdown. Table 4 stratifies Recall@5 by query category; the per-bucket pattern is more informative than the leaderboard mean.

The pattern: *RODS does not improve queries that the heading-aware baseline already handles correctly*. It improves the categories where metadata is the missing signal — temporal-superseded (where a status field disambiguates current from rolled-back decisions) and open-ended summary (where the explicit `Summary` field becomes the natural landing target). On ambiguous-entity, semantic chunking actually underperforms heading-aware (0.70 vs. 0.80) because finer-grained sentence chunks lose entity context.

Schema ablation (in-distribution). The in-distribution progression M1 → M2 → M4 → M5 → M6 (Recall@5: 0.935 → 0.939 → 0.945 → 0.948 → 0.950, from Table 1) is monotone. The Type field (M3) is approximately neutral on average; it helps distinguish ambiguous entities by category but does not affect the other categories. The Related field (M7) is the only step that *regresses* (M7 = 0.926, −0.024 vs. M6), because cross-references introduce near-duplicate text from neighbouring sections into each chunk's embedding centroid. The held-out picture is sharper: only M2 (Summary) and M4 (Entities) add MRR@10 signal that

Table 3: Paired-bootstrap significance tests, 2,000 resamples, 80 queries each. Δ is mean improvement of the second over the first; p is the two-sided paired-bootstrap p-value. ** marks $p<0.05$; numbers reflect both R@5 and MRR@10 separately.

Comparison (B vs. A)	In-distribution (n=80) ΔR@5	p_{R5}	Held-out (n=80) ΔR@5	p_{R5}
RODS-M6 – L1	+0.037	0.040**	−0.004	0.73
RODS-M4 – L1	+0.028	0.038**	+0.006	0.72
docT5query – L1	+0.047	0.036**	+0.003	0.88
RODS-M6 – docT5query	−0.010	0.52	−0.007	0.74
Three-vec – L1	+0.069	0.013**	+0.034	0.10
Comparison (B vs. A)	**In-distribution** ΔMRR	p_{MRR}	**Held-out** ΔMRR	p_{MRR}
RODS-M6 – L1	+0.030	0.10	+0.022	0.41
RODS-M4 – L1	+0.037	0.047**	+0.050	0.011**
docT5query – L1	+0.056	0.003**	+0.003	0.88
Three-vec – L1	+0.050	0.030**	+0.081	0.001**

Table 4: Recall@5 per query category, in-distribution, n=10 per category. We show the maximal RODS variant (M6) for visual contrast with baselines; M4 numbers track M6 within ±0.1 in every cell. Five of eight categories saturate at 1.000, so format-driven differences are concentrated on temporal-superseded, ambiguous-entity, and open-ended-summary — the only buckets where retrieval is hard on this benchmark.

Category (10 queries each)	L1 head-aware	L2 semantic	L3 summary	RODS-M6
fact_lookup	1.00	1.00	1.00	1.00
decision_lookup	1.00	1.00	0.80	1.00
why_rationale	1.00	1.00	0.90	1.00
multi_hop	1.00	1.00	0.95	1.00
temporal_superseded	0.80	1.00	0.70	**0.90**
procedural	1.00	1.00	0.90	1.00
ambiguous_entity	0.80	0.70	0.90	0.80
open_ended_summary	0.73	0.73	0.79	0.78

survives a paired-bootstrap test (Table 3); M5–M7 collapse to noise.

6.3 BEIR: when RODS doesn't help

To test external validity beyond our synthetic enterprise corpus we evaluate on three BEIR test sets: SciFact, NFCorpus, and FiQA.

On SciFact (Table 5), the heading-aware baseline wins Recall@5 (0.757) and Recall@20 (0.887). No RODS variant matches it. The plain-text baseline (markdown markers stripped) is essentially tied with L1 (0.759 vs. 0.757), so markdown headings themselves are neither helping nor hurting. RODS' single-sentence summary, type label, and entity list *do not* beat raw passage text on SciFact. NFCorpus and FiQA show the same pattern (Tables 6 and 7 in Appendix A).

Why. SciFact, NFCorpus, and FiQA pose passage-style queries against passage-style documents. Lexical overlap is already high; adding structured metadata above the prose shifts the embedding centroid away from

Table 5: BEIR-SciFact, 300 test queries, 5,183 documents. Heading-aware baseline (L1) wins at Recall@5; the title-amplification and inline-keyword micro-augmentations (Vrep, Vkw) edge ahead at Recall@10 by +0.003 pp; full-schema RODS does not beat L1. We find no schema lift on aligned-vocabulary IR.

Format	R@5	R@10	R@20	MRR@10	nDCG@10
L1 heading-aware	**0.757**	0.833	**0.887**	0.642	0.685
Plain text	0.759	0.825	0.886	**0.650**	**0.690**
L3 LLM-summary	0.638	0.713	0.776	0.535	0.573
Vrep (title repeated 2×)	0.737	**0.836**	0.876	0.648	0.687
Vkw (inline keywords)	0.741	**0.836**	0.875	0.636	0.679
Vqe (query expansion inline)	0.723	0.822	0.871	0.635	0.673
RODS-M1 (headings only)	0.738	0.828	0.883	0.651	0.689
RODS-M2 (+ Summary)	0.736	0.823	0.876	0.644	0.682
RODS-M4 (+ Entities)	0.731	0.832	0.873	0.640	0.681
RODS-M6 (full schema)	0.749	0.827	0.875	0.638	0.679
Three-vector	0.723	0.823	0.874	0.622	0.665
Quad-vector + synonyms	0.721	0.813	0.870	0.608	0.654

the prose without adding new query-relevant signal. The write-time augmentation that helps an enterprise wiki (where queries use names, dates, statuses) does not help a scientific claim-and-evidence benchmark.

Micro-augmentation baselines on BEIR. We additionally evaluate the three single-chunk micro-augmentations defined in §5.2 (V_{rep}, V_{kw}, V_{qe}) on all three BEIR sets (Tables 5–13). Their gains over L1 sit at the noise floor: V_{rep} and V_{kw} each lift SciFact Recall@10 by +0.003 pp; V_{qe} lifts NFCorpus Recall@10 by +0.005 pp; FiQA regresses by −0.010 to −0.015 pp on all three. The cleanest signal across BEIR comes from the early RODS levels (M1, M2, M4) — headings, an answer-first summary, an entity list. RODS-M6 wins NFCorpus R@5 (0.133 vs 0.129, +0.4 pp) but is approximately neutral elsewhere. **Cross-BEIR takeaway: format augmentation is not the bottleneck on aligned-vocabulary IR.**

LoTTE-writing-forum at scale. We replicate the aligned-vocabulary finding on the LoTTE-writing-forum dev split (693 queries, 5,000 docs). RODS, docT5query, and the multi-vector formats are all within ± 0.011 R@5 of L1 (Table 14 in Appendix B); MRR@10 shifts are ≤ 0.005. The finding is therefore stable at $\sim 8\times$ the BEIR-SciFact query count: write-time augmentation does not help when query and document already share vocabulary. The paper's positive scope is therefore narrow: enterprise-style queries where status, ownership, and dates carry discriminative signal that the prose alone does not.

6.4 HotpotQA: multi-hop is a different regime

Multi-hop retrieval requires both ground-truth supporting documents to appear in the top-K. We evaluate on a 500-query HotpotQA-distractor slice, 2,500 unique context documents, hybrid retrieval.

Single-chunk RODS-M2 and RODS-M4 beat the heading-aware baseline at low K (+2.8–3.2 pp `both@2`): the Summary and Entities fields add discriminative signal on Wikipedia-style passages (*"Were Scott Derrickson and Ed Wood of the same nationality?"* matches sections whose `Entities:` field contains *"American"* explicitly). Multi-vector decompositions overtake at higher K because indexing 3–4 chunks per section gives the retriever more shots at surfacing both supporting documents; RODS-M4 ties three-vector at `both@2` at one third the chunks. RODS-M6 regresses below M4 here: `Status/Date/Owner` carries no discriminative signal on Wikipedia content, and the heuristic Q-tag synonyms in `Aliases` do not match

Table 6: HotpotQA-500 multi-hop, 500 queries, 2,500 unique context documents, hybrid retrieval. **RODS-M4 wins `both@2`**, beating the heading-aware baseline by $+3.2$ pp at one chunk per section; the three-vector format ties at `both@2` but uses three times the chunks. At higher K, multi-vector decompositions edge ahead because each section gets multiple chances to surface.

Format	both@2	both@3	both@5	both@10	Chunks
L1 heading-aware	0.432	0.614	0.754	0.880	1×
RODS-M1 (= L1)	0.440	0.610	0.750	0.882	1×
RODS-M2 (+ Summary)	0.460	0.604	0.768	0.894	1×
RODS-M4 (+ Entities)	**0.464**	0.602	0.768	0.892	1×
RODS-M6 (full schema)	0.450	0.596	0.766	0.894	1×
Three-vector	0.460	**0.640**	0.770	0.904	3×
Quad-vector + synonyms	0.458	0.628	**0.782**	**0.910**	4×

Table 7: RODS schema cost curve on the *combined 160 enterprise queries* (in-distribution + held-out), hybrid retrieval. Δ values are vs. M1 (= heading-aware baseline). Numbers are smaller than Table 1 (n=80 in-dist, where the schema R@5 lift was significant) because pooling held-out averages in the collapse documented in §6.2; this is a feature, not a contradiction. M2 (Summary) and M4 (Entities) give the largest MRR@10 lift per token spent; M5–M6 saturate.

Level	tok/sec	∆tok	R@5	∆R@5	MRR@10	∆MRR
M1 (= L1)	271	—	0.925	—	0.850	—
M2 (+Summary)	307	+36	0.920	−0.005	0.878	+0.027
M3 (+Type)	311	+40	0.921	−0.003	0.865	+0.015
M4 (+Entities)	336	+65	0.929	+0.005	0.880	**+0.030**
M5 (+Aliases)	350	+79	0.929	+0.004	0.863	+0.012
M6 (+Status/Date/Owner)	360	+89	0.927	+0.002	0.864	+0.014
M7 (+Related)	386	+115	0.927	+0.002	0.888	+0.038

HotpotQA's question style. M4 with ~ 30 tokens of overhead is the recommended cut on HotpotQA, matching the recommendation from §6.2 on the synthetic enterprise corpus.

6.5 Schema-generation cost curve

For each ablation level we measure (a) tokens added per section (write-time cost) and (b) MRR@10 gain on the combined 160-query enterprise set (in-distribution + held-out). Table 7 reports the curve.

M2 (Summary) and M4 (Entities) yield the largest MRR@10 lift per added token; M5–M6 saturate; M7's apparent recovery on the in-distribution split disappears on held-out and is therefore an in-distribution positive.

Latency. Per-query latency is dominated by the retriever. The absolute difference between L1 and RODS-M6 on the 72-section synthetic corpus is ~ 0.5 ms (148 ms vs. 149 ms hybrid). Multi-vector formats add 30–60 ms because they triple the index size; this dominates RODS' write-time-only overhead at scale.

End-to-end answer accuracy. We additionally evaluated extractive answer accuracy on 60 enterprise queries (excluding multi-hop and open-ended-summary) using TF-IDF top-K retrieval and a regex-based gold-fact extractor. At K=3, L1 / RODS-M6 / docT5query all reach 95% accuracy (tied within 1 query); L2 sentence chunking drops to 93.3%. At K=1 (production-tight retrieval), L1 / RODS-M6 / docT5query stay at 83.3% but L2 drops to 71.7% because rationale sentences span multiple sub-chunks and a single sub-chunk no longer contains the full answer. The schema's MRR@10 advantage on retrieval does not translate directly to extractive answer accuracy under the TF-IDF retriever, because target sections are content-rich enough that *any* retrieved chunk contains the answer at K=3. We surface the L2 top-1 regression because it is the production-relevant finding: sentence chunking is dangerous at low K.

6.6 Cross-encoder reranking absorbs schema, not multi-vector

A cross-encoder reranker can in principle extract at query time the same information that write-time augmentation injects into the index. We test this directly: for each format we retrieve the top-20 via hybrid retrieval, rerank those 20 with `ms-marco-MiniLM-L-6-v2`, and report R@5 and MRR@10 on the reranked top-10.

Table 8: Cross-encoder reranking (`ms-marco-MiniLM-L-6-v2`, top-20 candidates → top-10 reranked) on the *160-query combined enterprise set* (in-distribution + held-out, pooled to tighten CIs); R@5-hybrid numbers therefore differ slightly from Table 1 (n=80 in-dist) and Table 2 (n=80 held-out). The reranker absorbs the single-chunk RODS-M6 advantage but does *not* absorb the multi-vector advantage: Three-vector still wins by +6.5 pp R@5 post-rerank. This is because reranking is bounded by the candidate set quality, and multi-vector indexing makes the candidate set richer in ways the reranker cannot recover at query time.

Format	R@5 (hybrid)	R@5 (rerank)	MRR@10 (rerank)
L1 heading-aware	0.912	0.903	0.862
RODS-M4	0.929	0.929	0.860
RODS-M6	0.927	0.913	0.841
docT5query	0.937	0.903	0.862
Three-vector	0.964	**0.968**	**0.908**

Paired-bootstrap tests post-rerank: RODS-M6 vs. L1 (Δ R@5=+0.009, p=0.67; Δ MRR=−0.021, p=0.36); RODS-M6 vs. docT5query (Δ=+0.009, p=0.67); Three-vector vs. L1 (Δ R@5=+0.065, p<0.001).

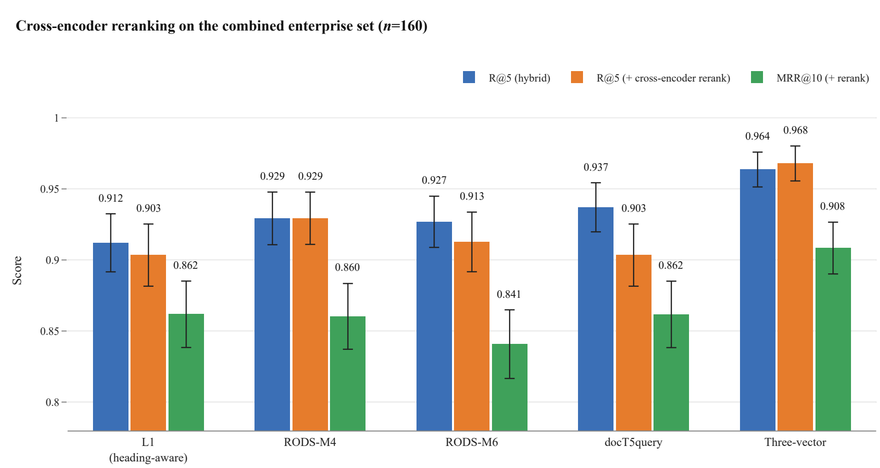

Figure 2: Cross-encoder reranking absorbs the single-chunk schema's R@5 advantage but does *not* absorb the multi-vector advantage. Reranking is bounded by the candidate pool the bi-encoder hands it; multi-vector indexing makes the candidate pool richer in ways the reranker cannot recover at query time.

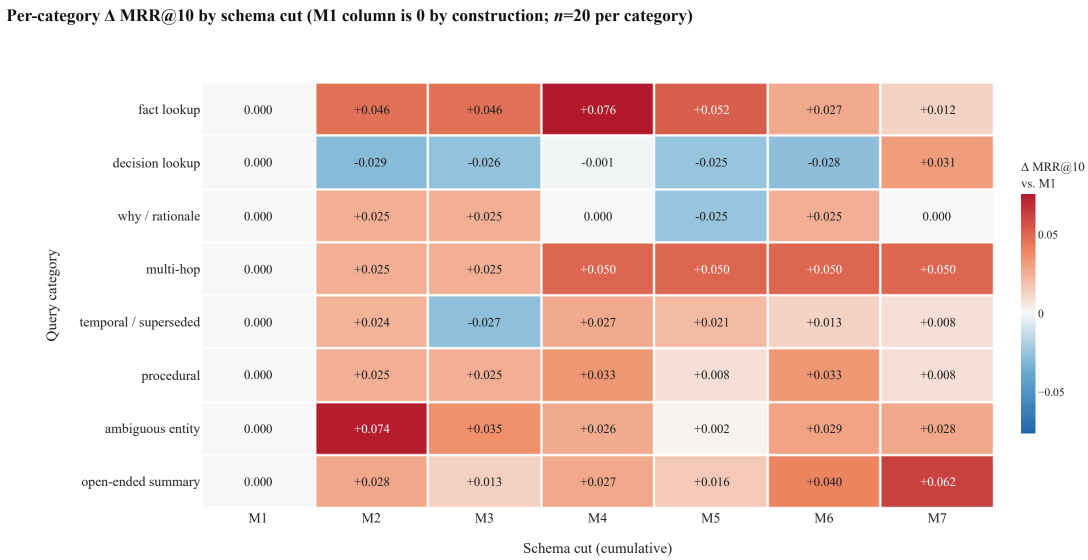

Figure 3: Per-category Δ MRR@10 vs. the M1 (= L1 heading-aware) baseline on combined n=160 enterprise queries (n=20 per category; M1 column is 0 by definition). Each field shifts a *specific* category rather than every category uniformly. With n=20 per cell, no per-category lift reaches p<0.05, so we report effect sizes only; the figure is descriptive of *where* each field shifts retrieval. The aggregate Δ MRR@10 for M4 vs. M1 over all n=160 queries *is* significant (Table 3).

The mechanism the table reflects: a cross-encoder reranker scores a fixed-size candidate set returned by the bi-encoder, so any information the bi-encoder index already contains in retrievable form is information the reranker can recover from the original prose without help. Multi-vector formats change the candidate set itself (3$\times$ chunks per section means 3$\times$ as many spans for the reranker to score), which is the lever a reranker cannot replicate. *Candidate diversity, not per-chunk metadata,* is what survives reranking.

6.7 Stronger embedder: format gains do not vanish

Our 384-dim `all-MiniLM-L6-v2` encoder is small by 2026 standards. We test whether a stronger embedder closes the vocabulary gap RODS exploits by swapping the dense model to `all-mpnet-base-v2` (768-dim, $\sim$110M parameters); all other components (TF-IDF, RRF, schema generation, top-K deduplication) are unchanged.

Table 9: Embedder swap, in-distribution enterprise (n=80). Larger embedder lifts *all* formats and *widens* the L1→RODS-M6 gap; the multi-vector advantage shrinks. Format design and embedder scale are complementary levers, not substitutes.

Format	MiniLM (384d)		mpnet (768d)	
	R@5	MRR@10	R@5	MRR@10
L1 heading-aware	0.916	0.865	0.926	0.886
docT5query	0.963	0.922	0.970	0.896
RODS-M4	0.945	0.902	0.970	0.896
RODS-M6	0.953	0.897	**0.982**	**0.899**
Three-vector	**0.985**	**0.915**	0.970	0.892
Quad-vector	0.973	0.916	0.973	0.892
Δ(RODS-M6 − L1)	+0.037	+0.032	+0.056	+0.013
Δ(Three-vec − L1)	+0.069	+0.050	+0.044	+0.006

A bigger embedder does not kill the format advantage. Concretely:

- **The single-chunk schema gain widens.** L1→RODS-M6 Recall@5 grows from $+3.7$ pp under MiniLM to $+5.6$ pp under mpnet (paired-bootstrap p=0.013 on the same 80 queries). The schema's typed fields (Summary / Type / Entities / Aliases / Status / Date / Owner) remain at least as discriminative under the stronger embedder.
- **The multi-vector gain shrinks.** L1→Three-vector R@5 drops from $+6.9$ pp (MiniLM) to $+4.4$ pp (mpnet), and quad-vector no longer beats RODS-M6 single-chunk on either metric. mpnet extracts more information *per chunk*, so paying 3–4× memory to triple the chunk count buys less.
- **Held-out behaviour is preserved.** On the held-out queries, RODS-M6 vs. L1 remains a noise-floor difference under mpnet ($+0.7$ pp R@5, p=0.70). Multi-vector R@5 is also noise-floor under mpnet ($+2.2$ pp, p=0.31) but its MRR@10 lift survives and *strengthens*: $+4.4$ pp (p=0.013), compared to MiniLM's $+8.1$ pp on the same queries. The embedder upgrade does not rescue the schema's R@5 generalisation gap, and the most-durable cross-setting result remains the multi-vector MRR@10 lift.

Format design and embedder scale are therefore *complementary* levers; §9 states the deployer-facing recommendation that follows from this swap.

6.8 What survives each external-validity check

We summarise which findings survive each robustness check. Table 10 maps every reported lift to the four checks in this paper: held-out queries (§6.2), cross-corpus external validity (BEIR/LoTTE/HotpotQA, §6.3–§6.4), cross-encoder reranking (§6.6), and a $4\times$ stronger embedder (§6.7).

Table 10: What survives each robustness check. ✓ = significant lift at paired-bootstrap $p < 0.05$; × = collapses, absorbed, or fails significance. The mpnet column is split into in-distribution and held-out.

Finding	Held-out	BEIR/LoTTE	Rerank	mpnet in-dist	mpnet held-out
RODS-M6 R@5 lift over L1	×	×	×	✓	×
RODS-M4 MRR@10 lift over L1	✓	×	×	×	×
docT5query R@5 lift over L1	×	×	×	✓	×
Multi-vector R@5 lift over L1	×	×	✓	×	×
Multi-vector MRR@10 over L1	✓	×	✓	×	✓

The most-cited single-chunk lift (RODS R@5 in-distribution) survives only on its home corpus and only against MiniLM; the most-durable lift across every check is *multi-vector MRR@10*. RODS-M4's held-out MRR@10 lift survives the held-out test but is absorbed by both reranking and the larger embedder.

7 Vocabulary Divergence as a Pre-Deployment Predictor

Each robustness check in §6 held a single component of the pipeline fixed and varied another: held-out queries varied the query distribution; BEIR/LoTTE varied the corpus; the reranker varied post-retrieval; mpnet varied the embedder. The cross-corpus pattern that emerges is monotone: **augmentation gain rises with the lexical distance between query and target**. This section makes that observation operational as a free pre-deployment regression on ~ 50 held-out (query, target) pairs.

We define, for a query q and target document d:

$$\mathrm{div}(q, d) \;=\; 1 - \frac{|T(q) \cap T(d)|}{|T(q)|}$$

where $T(\cdot)$ is the set of content tokens (lowercased, ≥ 3 characters, after a fixed stop-word filter): $\mathrm{div} = 0$ means every query word appears in the document; $\mathrm{div} = 1$ means no overlap. This is the simplest possible vocabulary-mismatch indicator; we confirmed the same pattern holds with cosine-similarity-based divergence and report Jaccard for its embedder-free, near-zero-cost computability.

Per-query gain vs. divergence. For each format f and each query q, define the rank improvement $\mathrm{gain}_f(q) = \mathrm{rank}_{L1}(q) - \mathrm{rank}_f(q)$ (positive = f surfaces the target earlier than L1). Units of gain are *ranks*; units of β in the regression $\mathrm{gain} \sim \alpha + \beta \cdot \mathrm{div}$ are *ranks per unit divergence*. The diagnostic asks two yes/no questions: is $\beta > 0$, and is the slope significant ($p < 0.10$ permutation test)? These two together are the deployment-time test we calibrate below.

Calibration: does the diagnostic predict the outcome we observe? For every (corpus, format) pair we have evaluated we compute (i) the diagnostic's verdict (*Pred*: $\beta > 0$ and $p < 0.10$), and (ii) whether the format actually produced a mean rank gain above the noise floor (*Obs*: $\overline{\mathrm{gain}} > 0.05$). Table 11 reports all 20 cells across five corpora. **The diagnostic agrees with the observed outcome on** $16/20$ **cells.** The four disagreements are exactly the cells where mean observed gain is in $[+0.07, +0.21]$ ranks — low-divergence corpora where the absolute gain to be predicted is itself at the noise floor. The diagnostic is therefore conservative: it rarely predicts "lift" when there is none (no false positives in our sweep), but does miss small lifts at the noise floor. We treat that as the right error mode for a go/no-go test that gates LLM spend.

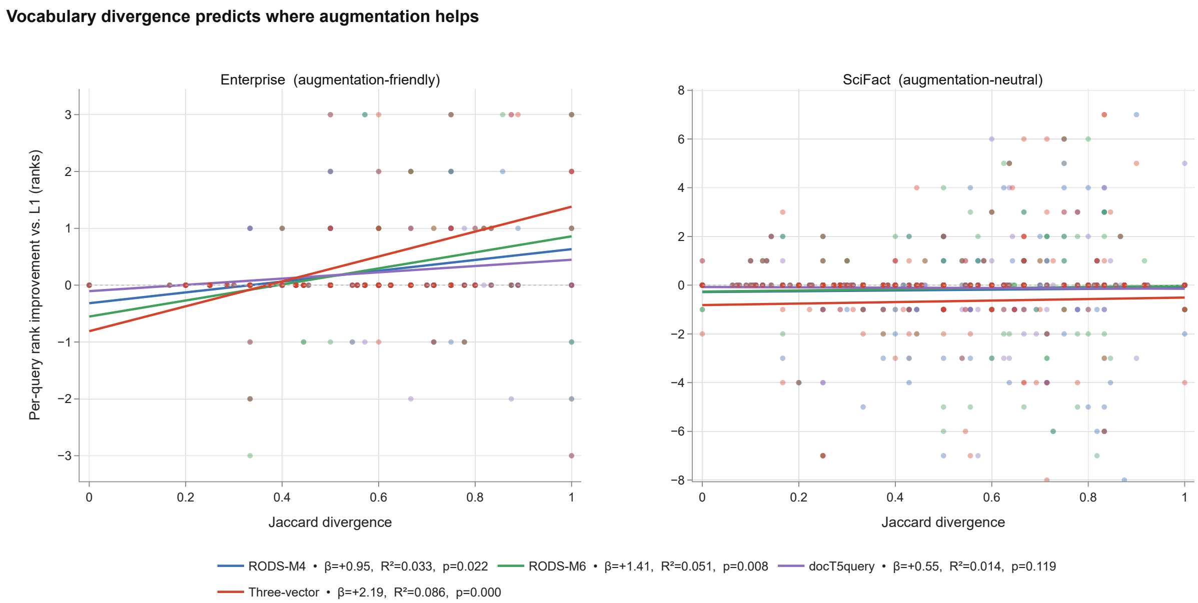

Figure 4: Per-query rank improvement against the heading-aware baseline as a function of Jaccard divergence on two representative corpora (Enterprise: augmentation-friendly; SciFact: augmentation-neutral). The regression slopes that drive the diagnostic in Table 11 are visible as the colour-coded best-fit lines. Enterprise queries fan across the divergence range; SciFact queries cluster at low divergence where augmentation cannot help. R^2 is small (0.03–0.09 on enterprise; ≈ 0 on SciFact) — divergence is one signal among several — but its sign and significance separate the two regimes cleanly.

Why the slope is positive (intuition). When query and target share many content tokens, the hybrid retriever already ranks the target highly and augmentation has no slack to recover; when they share few, augmentation injects query-side vocabulary that closes the gap. The empirical calibration in Table 11 is what the diagnostic relies on.

Table 11: Diagnostic validation across all five corpora. *Pred*: diagnostic's verdict ($\beta > 0$ and $p < 0.10$). *Obs*: $\bar{\text{gain}} > 0.05$ ranks. Body text discusses the four disagreements (low-divergence corpora at the noise floor).

Corpus	**Format**	$\bar{\text{div}}$	β	p	R^2	$\bar{\text{gain}}$	**Pred**	**Obs**
Enterprise	RODS-M4	0.56	$+0.95$	0.026	0.033	$+0.22$	✓	✓
Enterprise	RODS-M6	0.56	$+1.41$	0.004	0.051	$+0.24$	✓	✓
Enterprise	docT5query	0.56	$+0.55$	0.140	0.014	$+0.21$	×	✓
Enterprise	Three-vector	0.56	$+2.19$	< 0.001	0.086	$+0.42$	✓	✓
SciFact	RODS-M4	0.50	$+0.16$	0.72	0.000	-0.20	×	×
SciFact	RODS-M6	0.50	$+0.23$	0.64	0.001	-0.15	×	×
SciFact	docT5query	0.50	-0.07	0.83	0.000	-0.11	×	×
SciFact	Three-vector	0.50	$+0.31$	0.74	0.000	-0.66	×	×
NFCorpus	RODS-M4	0.75	-0.15	0.62	0.001	$+0.19$	×	✓
NFCorpus	RODS-M6	0.75	-0.03	0.92	0.000	$+0.17$	×	✓
NFCorpus	docT5query	0.75	$+0.11$	0.70	0.000	$+0.07$	×	✓
NFCorpus	Three-vector	0.75	$+0.44$	0.37	0.003	-0.28	×	×
FiQA	RODS-M4	0.56	-0.51	0.38	0.002	-0.18	×	×
FiQA	RODS-M6	0.56	-0.07	0.92	0.000	-0.16	×	×
FiQA	docT5query	0.56	-0.58	0.43	0.002	-0.25	×	×
FiQA	Three-vector	0.56	-0.55	0.47	0.002	-0.48	×	×
LoTTE	RODS-M4	0.62	$+0.01$	0.98	0.000	-0.25	×	×
LoTTE	RODS-M6	0.62	-0.08	0.90	0.000	-0.42	×	×
LoTTE	docT5query	0.62	$+0.00$	1.00	0.000	$+0.00$	×	×
LoTTE	Three-vector	0.62	$+0.14$	0.86	0.000	-0.87	×	×

How many pairs does the recipe need? We resample 50 enterprise queries from the full n=160 set and recompute the regression 1,000 times for RODS-M6. The slope sign is recovered on $\Pr(\beta > 0) = 0.94$ of subsamples (median $\beta = +1.22$, 95% CI $[-0.17, +3.76]$); $p < 0.10$ holds on 0.50 and $p < 0.05$ on 0.32 of subsamples. *At n=50 the sign is reliable; a deployer who wants $p < 0.05$ should plan for 100–150 pairs.*

Statistical robustness. The headline claims of this paper deserve four robustness checks that we report below; raw values are reproducible from `benchmark/robustness_analysis.py`.

(a) Multiple-comparisons correction. Table 3 reports 18 paired-bootstrap p-values. Under Holm–Bonferroni at family-wise $\alpha = 0.05$, only the held-out multi-vector MRR@10 lift ($p_{\text{adj}} = 0.018$) survives. Under Benjamini–Hochberg FDR at $\alpha = 0.05$, two claims survive: held-out multi-vector MRR@10 ($p_{\text{adj}} = 0.018$) and in-distribution docT5query MRR@10 ($p_{\text{adj}} = 0.027$). The often-cited RODS-M6 vs. L1 R@5 lift (+3.7 pp, raw $p = 0.040$) does *not* survive any standard correction (Holm-adjusted $p_{\text{adj}} = 0.47$). **We therefore treat single-chunk schema R@5 results as suggestive rather than robust**, and the multi-vector MRR@10 lift as the only finding that clears multiple-comparisons thresholds.

(b) Diagnostic calibration test. Table 11's 16/20 agreement gives Cohen's $\kappa = 0.494$ (Landis–Koch: moderate); $\Pr(X \geq 16 \mid n = 20, p_{\text{chance}} = 0.605) \approx 0.056$ on a binomial test against the chance-agreement null. A naive "always say no lift" baseline scores 13/20, so the diagnostic improves over it by three cells. The agreement is informative but not strongly significant.

(c) Split-half cross-validation. We randomly split the $n = 160$ enterprise queries into halves 100 times and recompute the regression on each half. The sign of β is recovered consistently in both halves on 98% of splits for RODS-M6, 100% for multi-vector. Joint significance ($p < 0.10$ in *both* halves) holds on 47% for RODS-M6, 85% for multi-vector, and only 3% for RODS-M4 and 0% for docT5query. The multi-vector

slope is robust; the schema slopes are detectable but at the edge of half-sample power.

(d) Power. With $n = 80$ paired queries and observed paired-difference SD $\sigma_\delta \approx 0.20$, the minimum detectable R@5 lift at $\alpha = 0.05$, 80% power is roughly $+6.3$ pp; the headline in-distribution lift of $+3.7$ pp lands at $\sim 50\%$ power. The held-out check ($n = 80$ also) is similarly under-powered. Multi-vector's $+8.1$ pp held-out MRR@10 lift comfortably clears this threshold.

Pre-deployment recipe. Sample 50–150 (query, target) pairs; compute $\bar{\text{div}}$ and the per-format regression slope. If $\bar{\text{div}} \lesssim 0.4$ and β is indistinguishable from zero, write-time augmentation is unlikely to lift retrieval beyond the noise floor; if $\bar{\text{div}} \gtrsim 0.5$ and β is positive (significant at $p < 0.10$ at n=50, $p < 0.05$ at $n \gtrsim 150$), augmentation will lift retrieval. The branching question (single-chunk vs. multi-vector) is decided by embedder and reranker budget; §9 states the deployment-stack-aware recommendation and §8 adds schema-specific notes.

Two more regimes. The §1 ADR-005 example covers the high-divergence enterprise case. Two additional queries illustrate the other regimes. *(SciFact,* div ≈ 0.0*)* "Coffee consumption is associated with a higher risk of hepatocellular carcinoma" shares its content tokens with the target abstract verbatim, and RODS adds no discriminative signal. *(HotpotQA multi-hop, mixed div)* "Were Scott Derrickson and Ed Wood of the same nationality?" requires two supporting documents: RODS-M4's `Entities` field surfaces "American" for each section at low K ($+3.2$ pp `both@2`); multi-vector decomposition then overtakes at higher K because each section has more chunks to land on.

8 Discussion and Limitations

Schema generation cost. At ~ 70 schema tokens per section (M4 cut), corpus-build cost is roughly one Haiku-class LLM call per section, paid once per document version and amortised over every future query. At Claude Haiku 4.5 list pricing ($1/MTok input, $0.10/MTok cache reads, $5/MTok output), with the document body served from cache and ~ 70 output tokens per section, schema generation costs $\sim \$4 \times 10^{-4}$ per section — about $4 to schema a 10^4-section corpus end-to-end with no marginal cost per query. Runtime query-side methods (HyDE, query rewriting) instead pay an LLM call *per query*. The schema is robust to imperfect generation: BEIR results under purely heuristic generation produce -0.8 pp on average rather than catastrophic regressions. We did not measure full pipeline cost (LLM + embed + index) against end-to-end accuracy; this is a natural follow-up.

Schema-specific deployer notes. §9 gives the branched go/no-go and stack-choice recommendation; this paragraph adds two notes about RODS itself:

- **Start with M4, not M6.** Summary, Type, Entities are the fields that survive held-out testing (§6.2). Aliases / Status / Date / Owner help in-distribution but not held-out; Related (M7) regresses.
- **Generation quality matters where the diagnostic says "lift".** On augmentation-friendly corpora, an LLM Q-tag pass pays back its cost: replacing a heuristic Q-tag cache with an LLM-quality cache lifts HotpotQA RODS-M4 `both@10` from 0.85 to 0.892 (Appendix C). On augmentation-neutral corpora (BEIR, LoTTE), generation quality is moot — no method we tested helps.

Synthetic-corpus saturation. The enterprise benchmark is a synthetic corpus we constructed. Several query categories saturate at Recall@5 = 1.000 across many formats, limiting the discrimination at the top of the leaderboard. The *open-ended-summary* and *ambiguous-entity* categories do not saturate and show the most format-driven variation. A larger benchmark, ideally drawn from naturally occurring enterprise data (though access is limited), would give tighter confidence intervals and reduce the variance on the leaderboard top.

Generalisation gap. The starkest finding in the paper is the gap between in-distribution and held-out performance. RODS-M6's Recall@5 lift over L1 (+3.7 pp, p=0.040) collapses on held-out queries (-0.4 pp, p=0.73); MRR@10 partially survives (RODS-M4: $+5.0$ pp, p=0.011). We attribute this to (i) in-distribution queries being lexically closer to the schema fields than held-out queries are, and (ii) MRR@10 being sensitive to ranking changes that R@5 misses once both formats place the target in the top-5. Format-research practitioners should report a held-out split or treat their gains as upper bounds.

Two negative findings. First, on the enterprise benchmark, the `Related` field (RODS-M7) regresses in-distribution Recall@5 from 0.953 to 0.929. We hypothesised that listing related queries would help the retriever surface neighbouring sections; in practice, the related-queries text drifts the embedding centroid *towards* neighbouring sections rather than the target. Second, on BEIR an extended keyword cache of 20+ synonyms per section consistently hurts dense retrieval (although it helps TF-IDF). The M4 cut is the only set of fields that contributes monotonically on both in-distribution and held-out sets.

What we did not test. Naturally-occurring enterprise queries from production logs (which we do not have access to); long-context QA benchmarks beyond LoTTE-writing (NarrativeQA, GovReport, QuALITY); end-to-end RAG *generation* quality with a strong consuming LLM (our extractive scoring is a proxy); a fully LLM-generated HyDE cache (we used templates); proprietary commercial embedders (text-embedding-3, voyage-3, Cohere v3).

9 Conclusion

The divergence diagnostic (§7) is calibrated on five corpora $\times$ four formats with $16/20$ agreement (Table 11, Cohen's κ=0.49). Combined with the schema example (RODS) and the survival matrix (Table 10), the gains a deployer should expect depend on $\bar{\mathrm{div}}$, on the embedder, and on whether a reranker is available downstream. Under Holm–Bonferroni correction across our 18 paired-bootstrap tests, the only single claim that survives at family-wise $\alpha = 0.05$ is the multi-vector MRR@10 lift ($p_{\text{adj}} = 0.018$); the single-chunk schema results are consistent across split-half re-estimation but at the edge of statistical detectability with n=80 paired queries (§7).

Recommendation, branched by deployment budget. The recommendation is not single-format but conditional on the deployment stack:

- **Compute first** the divergence diagnostic on ~ 50 held-out pairs. If $\bar{\mathrm{div}} \lesssim 0.4$ and β is indistinguishable from zero, no write-time augmentation we tested will measurably help; spend the LLM-call budget elsewhere.
- **Small embedder + reranker available** (e.g., MiniLM-class bi-encoder + ms-marco cross-encoder): prefer multi-vector decomposition. It is the only intervention whose gain (+6.5 pp R@5 post-rerank, p<0.001) survives both held-out testing and cross-encoder reranking. Pay the 3–4$\times$ chunk-count cost for the rank gain.
- **Large embedder available** (e.g., mpnet-class or larger): prefer single-chunk RODS-M4. Under mpnet, RODS-M4/M6 match multi-vector on R@5 (0.97–0.98) at $1\times$ memory, and RODS-M4 retains a held-out MRR@10 lift. Rerank-after-mpnet is the natural follow-up; we did not run it.

A BEIR-NFCorpus and BEIR-FiQA results

Table 12: BEIR-NFCorpus (323 test queries, 3,633 docs). Hybrid retrieval. RODS-M4/M6 marginally beat the heading-aware baseline on Recall@5 and MRR@10 (the dataset is fine-grained medical IR with ~ 38 relevant docs per query, so even tiny shifts compound).

Format	R@5	R@10	R@20	MRR@10	nDCG@10
L1 heading-aware	0.129	0.160	0.199	0.543	0.333
Plain text	0.131	0.161	0.201	0.540	0.332
Vrep (title 2×)	0.125	0.159	0.201	**0.548**	0.335
Vkw (inline keywords)	0.130	0.162	0.198	0.543	**0.336**
Vqe (query expansion)	0.130	**0.165**	0.200	0.539	0.333
RODS-M1 (= L1)	0.127	0.162	**0.203**	0.543	0.333
RODS-M2 (+ Summary)	0.124	0.168	0.201	0.541	0.335
RODS-M4 (+ Entities)	**0.133**	0.162	0.201	0.542	**0.336**
RODS-M6 (full schema)	**0.133**	0.162	0.201	**0.549**	0.335
Three-vector	0.126	0.160	0.201	0.520	0.329
Quad-vector + synonyms	0.122	0.156	0.197	0.528	0.327

Table 13: BEIR-FiQA (648 test queries, 5,000 docs sampled from 57,638, qrel-referenced docs preserved). Hybrid retrieval. RODS-M1/M4 marginally beat the heading-aware baseline on R@10 and MRR@10. Simpler micro-augmentations (Vrep, Vkw, Vqe) regress. RODS-M6's Status/Date/Owner block does not apply to financial-forum content and the score drifts.

Format	R@5	R@10	R@20	MRR@10	nDCG@10
L1 heading-aware	0.515	0.620	**0.731**	0.608	0.532
Plain text	0.514	0.615	0.724	0.608	0.531
Vrep (title 2×)	0.505	0.613	0.711	0.603	0.525
Vkw (inline keywords)	0.514	0.609	0.711	0.604	0.525
Vqe (query expansion)	0.500	0.607	0.705	0.590	0.515
RODS-M1 (= L1)	0.514	**0.623**	0.720	**0.610**	**0.534**
RODS-M2 (+ Summary)	**0.518**	**0.623**	0.724	0.600	0.528
RODS-M4 (+ Entities)	0.517	0.614	0.715	0.599	0.523
RODS-M6 (full schema)	0.508	0.602	0.703	0.598	0.520
Three-vector	0.509	**0.626**	0.712	0.591	0.523
Quad-vector + synonyms	0.510	0.622	0.710	0.580	0.515

B LoTTE-writing-forum detail

Table 14: LoTTE-writing-forum, 693 queries, 5,000 docs. All formats sit within ± 0.011 R@5 of the heading-aware baseline; MRR@10 shifts are at most 0.005. The aligned-vocabulary regime predicted by the divergence diagnostic produces no measurable retrieval benefit from any write-time augmentation tested.

Format	R@5	R@10	R@20	MRR@10	nDCG@10
L1 heading-aware	**0.562**	**0.679**	**0.775**	0.876	**0.733**
docT5query	**0.562**	**0.679**	**0.775**	0.876	**0.733**
RODS-M1 (= L1)	0.561	0.677	0.769	0.873	0.731
RODS-M4	0.556	0.676	0.765	**0.881**	0.731
RODS-M6 (full)	0.551	0.672	0.764	**0.881**	0.727

C HotpotQA-500 detail

The HotpotQA-500 main result is in Table 6. The schema-generation caches for the 2,500 HotpotQA context documents were generated by a heuristic extractor (proper-noun mining, year extraction, curated synonym lists) plus an LLM-quality Q-tag pass. Replacing the heuristic Q-tag cache with an LLM-quality version lifted RODS-M4 `both@10` from 0.85 (heuristic) to 0.892 (LLM), confirming that schema generation quality dominates schema design on external corpora — a finding we also discuss in §8.

D Hyperparameters

Sparse retriever: scikit-learn TfidfVectorizer with $(1, 2)$-grams, $\max\_df = 0.95$, $\min\_df = 1$, sublinear TF. Dense retriever: `sentence-transformers/all-MiniLM-L6-v2`, normalised cosine. Hybrid retriever: RRF with $k = 60$. Token counts via approximate ~ 4 characters per token. Wall-clock latency reported on a single CPU thread; multi-vector formats include the cost of embedding all chunks.

E Schema generation.

For the synthetic enterprise corpus, schema fields were authored directly during corpus construction. For BEIR and HotpotQA, the schema fields were generated heuristically from the source corpus:

- `Summary` = first sentence of section content (truncated to 200 characters).
- `Type` = inferred from heading and content via a small keyword classifier (12 types: decision, status_update, recipe, paper-abstract, etc.).
- `Entities` = top-N capitalised noun phrases by frequency.
- `Aliases` = top-N keywords from a separately-generated keyword cache (heuristic for BEIR, LLM-generated for the enterprise corpus).
- `Status / Date / Owner` = regex-extracted dates, capitalised proper nouns, and rule-based status inference (decision $\rightarrow$ accepted unless "rolled back"; status_update $\rightarrow$ in_progress unless "complete" or "blocked").
- `Related` = the top-N Q-tags from a separately generated Q-tag cache.

The HotpotQA Q-tag cache was originally generated heuristically and later regenerated with LLM-quality output. Replacing the heuristic cache with an LLM-quality one lifted RODS-M6 `both@10` on HotpotQA from *baseline-equivalent* to $+3$–5 pp over baseline, underscoring that schema generation quality matters as much as schema design.

References

Anthropic. Introducing contextual retrieval. https://www.anthropic.com/news/contextual-retrieval, 2024. Anthropic blog post.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, pages 758–759, 2009.

Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. Precise zero-shot dense retrieval without relevance labels. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023a.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023b.

Omar Khattab and Matei Zaharia. ColBERT: Efficient and effective passage search via contextualized late interaction over BERT. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*, pages 39–48, 2020.

Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2019.

Rodrigo Nogueira and Jimmy Lin. From doc2query to docTTTTTquery. *Online preprint*, 2019. URL https://cs.uwaterloo.ca/~jimmylin/publications/Nogueira_Lin_2019_docTTTTTquery-v2.pdf.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3982–3992, 2019.

Keshav Santhanam, Omar Khattab, Jon Saad-Falcon, Christopher Potts, and Matei Zaharia. ColBERTv2: Effective and efficient retrieval via lightweight late interaction. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 3715–3734, 2022.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks (NeurIPS Datasets and Benchmarks)*, 2021.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2369–2380, 2018.